CDO Risk Simulation and Waterfall Analysis

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Methodology

We modeled a five-year collateralized debt obligation backed by ten identical speculative-grade corporate bonds, each with $10 million face value, a 6% annual coupon paid quarterly, a 4% annual default probability, and 60% loss given default. A constant 0.20 pairwise dependence parameter was imposed through a 10 x 10 Gaussian correlation matrix. Using a fixed random seed, 1,000 sets of independent standard-normal draws were transformed with a Cholesky factor to generate correlated normals. The standard-normal CDF then mapped these draws to correlated uniforms, which were converted to default times using the constant annual default-probability relationship.

Bond cash flows were generated quarterly. Before default, each surviving bond paid its scheduled coupon; in the default quarter it paid recovery equal to 40% of face value and no coupon; after default, no further cash flow was paid. Bonds surviving through quarter 20 paid both the final coupon and face value. Collateral interest was allocated first to Class A interest, then Class B interest, with the residual paid to equity. Recovery and principal proceeds were applied sequentially to Class A principal, then Class B principal, and finally equity. Tranche coupons were calculated on beginning-of-quarter outstanding principal balances. This principal allocation is a modeling assumption because the project description does not specify a detailed amortization schedule.

Table 1. Model Parameters

Input	Base Case
Number of simulations	1,000
Number of bonds	10
Bond face value	$10MM each
Maturity	5 years / 20 quarters
Annual coupon	6% paid quarterly
Annual default probability	4%
LGD / recovery	60% / 40%
Default correlation parameter	0.20
Class A	$20MM notional; 2% coupon
Class B	$10MM notional; 4% coupon

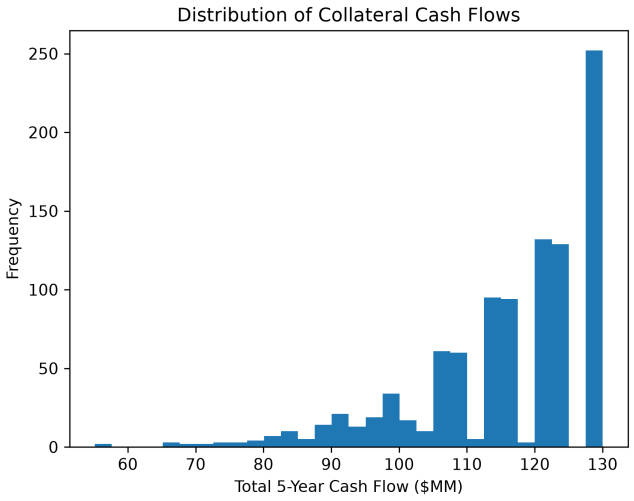
Table 2. Base-Case Five-Year Cash-Flow Results

Component	Mean ($MM)	Std. Dev. ($MM)	5th Pctl. ($MM)	Min ($MM)	Max ($MM)
Collateral	115.94	13.57	89.79	55.15	130.00
Class A	21.65	0.39	20.80	20.16	22.00
Class B	11.99	0.08	12.00	10.76	12.00
Equity	82.31	13.17	57.05	24.23	96.00

Base-Case Cash-Flow Distributions

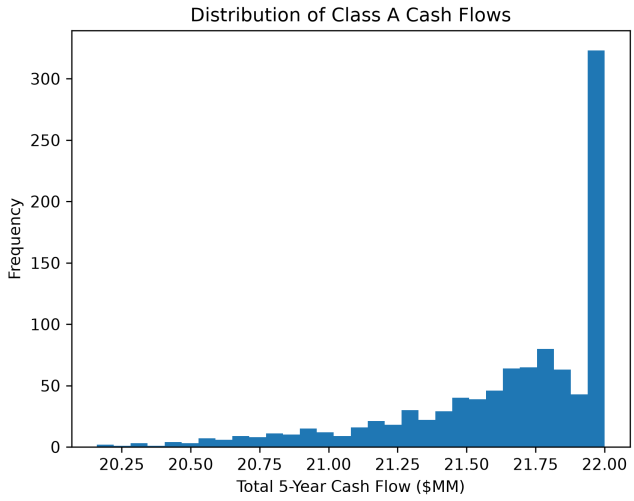
The collateral distribution reflects the effect of correlated defaults on total five-year cash flow. Class A is much more tightly distributed because it receives first priority in the waterfall.

Figure 1. Distribution of Collateral Cash Flows



Total five-year collateral cash flow across 1,000 correlated default scenarios.

Figure 2. Distribution of Class A Cash Flows

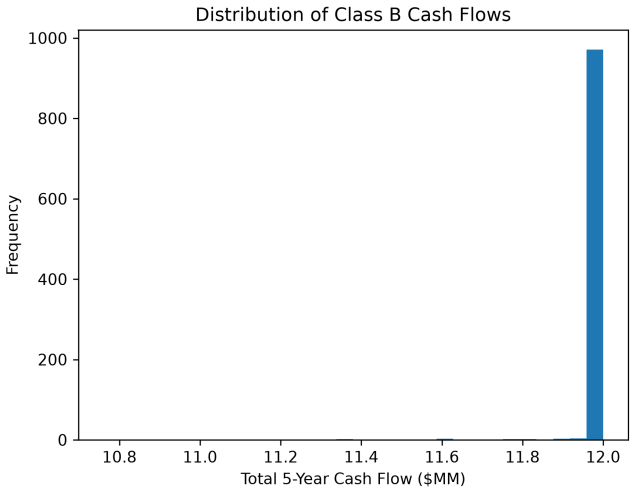


Class A is concentrated near its maximum because of senior payment priority and subordination.

Tranche and Equity Distributions

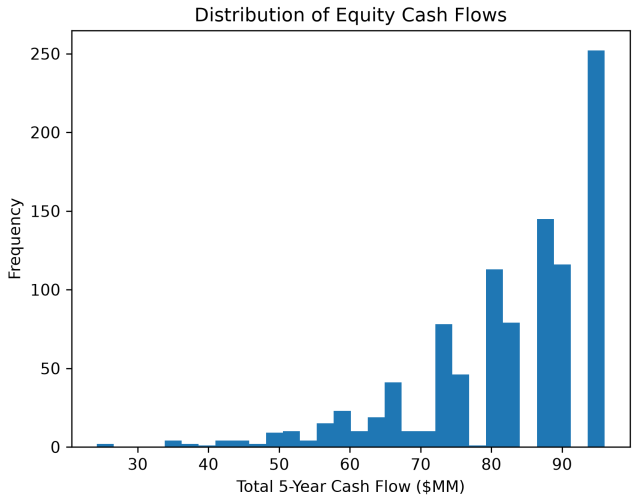
Class B remains strongly protected in the base simulation sample, while equity absorbs most of the variation in collateral performance. The contrast between the two distributions illustrates how subordination transfers credit variability to the residual tranche.

Figure 3. Distribution of Class B Cash Flows



Class B is highly concentrated around $12MM in the base sample.

Figure 4. Distribution of Equity Cash Flows



Equity exhibits the broadest downside distribution because it absorbs residual collateral risk.

Sensitivity Analysis

Sensitivity tests varied default probability, loss given default, and default correlation one at a time while holding the random seed fixed. Higher PD and LGD reduce both expected cash flows and lower-tail outcomes. Correlation has comparatively little effect on the mean but materially changes dispersion and downside risk because it changes the degree to which defaults cluster across obligors.

Table 3. Sensitivity Results (USD millions)

Scenario	Coll. Mean	Coll. Std	Coll. 5%	Eq. Mean	Eq. Std	Eq. 5%
Base	115.94	13.57	89.79	82.31	13.16	57.05
Low PD	122.90	9.49	103.86	89.08	9.23	70.84
High PD	110.03	15.75	80.64	76.57	15.24	48.44
Low LGD	119.64	10.11	99.79	86.21	9.46	68.03
High LGD	112.25	17.03	79.79	78.43	16.82	46.42
Low Corr	116.06	11.62	93.08	82.42	11.29	59.90
High Corr	116.21	16.78	80.63	82.58	16.24	48.30

Figure 5. Collateral Tail Sensitivity

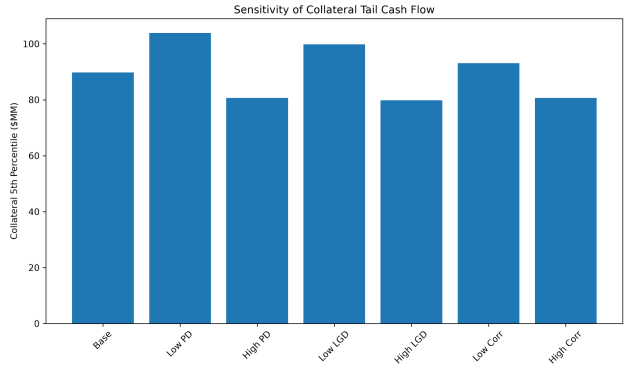
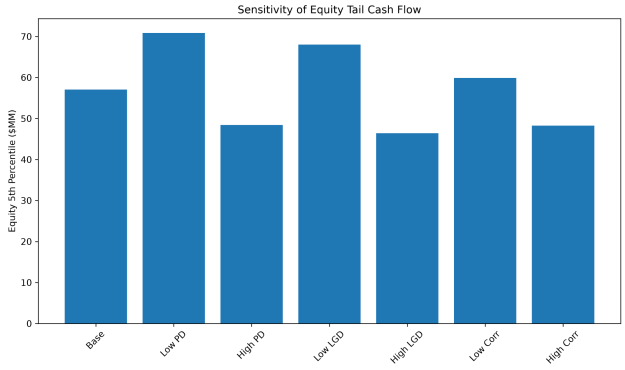


Figure 6. Equity Tail Sensitivity



Fifth-percentile cash flow by scenario. Higher default correlation leaves the mean nearly unchanged but materially worsens the lower tail.

Conclusion

The base-case simulation produces mean five-year collateral cash flow of $115.94MM, with mean Class A, Class B, and equity cash flows of $21.65MM, $11.99MM, and $82.31MM, respectively. No Class A or Class B interest or principal shortfall was observed in the 1,000 base paths under the stated waterfall assumptions. Sensitivity results show that PD primarily shifts expected default losses, LGD changes loss severity, and correlation mainly amplifies joint downside outcomes. The model therefore provides a reproducible Part 1 framework for evaluating CDO cash-flow risk before discounted valuation is introduced.

Mini-Project 3B: CDO Cash-Flow Simulation & Lintner Model Valuation

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Executive Summary

This report values the three tranches of the simulated collateralized debt obligation (CDO) using the Lintner mean-variance framework discussed in class. The collateral pool consists of ten five-year corporate bonds with quarterly cash flows, 4% annual default probability, 60% loss-given-default, and 20% default correlation. Part II resets the liability structure to Class A principal of $15 MM at a 4.5% target coupon, Class B principal of $30 MM at a 5.5% target coupon, and the remaining $55 MM as residual equity. All scenario cash flows are discounted at the 4% risk-free rate. The original project assumption of a 9% bond yield-to-maturity (YTM) is retained for market valuation of the collateral pool and for calibration of the aggregate Lintner risk charge.

Using 1,000 fixed moment-matched simulations, the expected risk-free present value of the collateral pool is $97.7297 MM, while its market value at a 9% YTM is $88.0272 MM. The resulting aggregate risk charge is $9.7025 MM. The Lintner values are $15.1982 MM for Class A, $31.9771 MM for Class B, and $40.8519 MM for the residual. Their sum equals the $88.0272 MM market value of the collateral pool, providing an internal valuation consistency check. In the bonus optimization, the largest Class B principal with zero observed Class B payment defaults across the 1,000 fixed scenarios is $27.90 MM.

Step 1: Scheduled Zero-Default Waterfall & Risk-Free Discounting

The scheduled waterfall is first computed under a zero-default scenario. With ten $10 MM bonds paying a 6% annual coupon quarterly, the collateral generates $1.50 MM per quarter in Q1-Q19 and $101.50 MM in Q20. Class A receives $0.16875 MM per quarter plus $15.00 MM principal in Q20, while Class B receives $0.41250 MM per quarter plus $30.00 MM principal in Q20. The residual receives all remaining scheduled collateral cash flow.

Period	Collateral ($MM)	Class A ($MM)	Class B ($MM)	Residual ($MM)
Q1-Q19 (each quarter)	1.5000	0.1688	0.4125	0.9188
Q20	101.5000	15.1688	30.4125	55.9188
Scheduled PV @ 4%	109.0228	15.3384	32.0301	61.6543

Step 2: Moment-Matched Simulation Engine

Because only 1,000 simulations are used, the static standard-normal draws are moment matched before Cholesky correlation is applied. Each independent normal column is standardized to a sample mean of zero and a sample standard deviation of one; the correlated normal columns are then re-standardized in the same way. This reduces avoidable Monte Carlo sampling noise while preserving the fixed random-number framework used for valuation and structure comparisons.

- Matched independent normal means: approximately 0.0000 for all ten bonds.
- Matched independent normal standard deviations: 1.0000 for all ten bonds.
- Correlated normal standard deviations after re-standardization: 1.0000 for all ten bonds.

Step 3: Scenario PVs, Covariance Matrix, and Default Indicators

For each simulation, the realized Class A, Class B, and residual cash flows are discounted at 4% per annum, using 1% per quarter. The mean scenario PVs are $15.2634 MM for Class A, $32.0146 MM for Class B, and $50.4517 MM for the residual. These sum to an expected collateral PV of $97.7297 MM.

Tranche	Notional ($MM)	Mean PV @ 4% ($MM)	Default / Shortfall Freq.	Comment
Class A	15.00	15.2634	0.00%	No interest or principal shortfall
Class B	30.00	32.0146	0.30%	3 of 1,000 scenarios with interest shortfall
Residual	55.00	50.4517	n/a	First-loss / residual claim
Total Pool	100.00	97.7297	75.30%*	*At least one collateral bond defaults

The covariance matrix of the tranche PVs, in $MM², is shown below. The residual absorbs nearly all portfolio variance, while Class A and Class B have very small stand-alone variance and covariance contributions.

Covariance ($MM²)	Class A	Class B	Residual
Class A	0.0060	0.0028	0.7683
Class B	0.0028	0.0041	0.4403
Residual	0.7683	0.4403	113.2402

The covariance of each tranche with the total collateral pool is the row sum of the covariance matrix: 0.7771 $MM² for Class A, 0.4472 $MM² for Class B, and 114.4488 $MM² for the residual. The total portfolio variance is 115.6730 $MM². This confirms that the residual carries almost all systematic collateral risk, while the two debt tranches are strongly protected by subordination.

Step 4: Implicit Cost of Risk from the Underlying Bonds

The Lintner framework requires a market risk charge for the collateral portfolio. The project assumptions specify a 9% YTM for the underlying corporate bonds, so the simulated bond cash flows are valued using a 9% market discount rate. The resulting market value is $8.8027 MM per bond, or $88.0272 MM for the ten-bond collateral pool.

Total Risk Charge = E[PV at RF] - Market Value = 97.7297 - 88.0272 = 9.7025 MM

Dividing the aggregate risk charge by the total portfolio variance gives the price of variance risk used to allocate the Lintner charge across tranches: 0.083878 per $MM² of variance. Risk is then allocated in proportion to each tranche's covariance with the total pool.

Calibration Item	Value
Expected collateral PV @ 4% RF	$97.7297 MM
Collateral market value @ 9% YTM	$88.0272 MM
Aggregate risk charge	$9.7025 MM
Portfolio variance	115.6730 $MM²
Cost of risk	0.083878 per $MM² of variance

Step 5: Lintner Tranche Valuation

The Lintner value of each tranche equals its mean risk-free PV minus a covariance-based risk charge. The debt tranches receive very small risk deductions because their covariance with the total pool is small. The residual receives nearly the entire aggregate risk deduction.

Tranche	Mean PV @ RF ($MM)	Cov. w/ Pool ($MM²)	Risk Charge ($MM)	Lintner Value ($MM)	Value / Par
Class A	15.2634	0.7771	0.0652	15.1982	101.32%
Class B	32.0146	0.4472	0.0375	31.9771	106.59%
Residual	50.4517	114.4488	9.5998	40.8519	74.28%
Total CDO	97.7297	115.6730	9.7025	88.0272	88.03% of par

The sum of tranche values is exactly $88.0272 MM, equal to the market value of the underlying bond pool. This equality is the key internal consistency check for the Lintner allocation: the model redistributes the collateral portfolio risk premium across tranches without creating or destroying aggregate value.

Step 6 (Bonus): CDO Capital Structure Optimization

For the bonus analysis, Class A principal is held fixed at $15 MM and Class B principal is varied while the residual absorbs the offsetting change in subordination. The same 1,000 fixed moment-matched scenarios are reused at every structure, so differences across structures reflect only the liability design rather than new random draws.

A scenario-level Class B default indicator is set to one if Class B experiences any interest shortfall or has unpaid principal remaining at maturity. Under the original $30 MM Class B structure, the observed Class B default frequency is 0.30%. A fine grid search around the zero-default boundary shows that $27.90 MM is the largest Class B principal with no observed Class B payment default in the 1,000 fixed scenarios; at $27.91 MM, the observed default frequency rises to 0.30%.

We want to maintain a zero observed default rate on Class B because institutional CDO structuring requires debt tranches to maintain a pristine credit profile to satisfy investment-grade rating constraints. Any debt tranche with a noticeable default rate above 0% risks being categorized as speculative grade ('junk'), which would legally bar or deter institutional investors, such as pension funds, insurance companies, and bank treasuries, from purchasing the tranche.

Structure	Class B Principal ($MM)	Class B Lintner Value ($MM)	Residual Lintner Value ($MM)	Observed B Default Freq.
Original structure	30.00	31.9771	40.8519	0.30%
Max zero-observed-default structure	27.90	29.7353	43.0937	0.00%

The optimization figure below shows the trade-off among residual value, Class B market premium over par, and the default-indicator frequency as Class B principal increases. At low and moderate leverage, Class B can command a premium because its 5.5% target coupon exceeds the 4% risk-free rate. At higher leverage, increasing credit losses and covariance risk erode that premium and cause the default indicator to rise sharply.

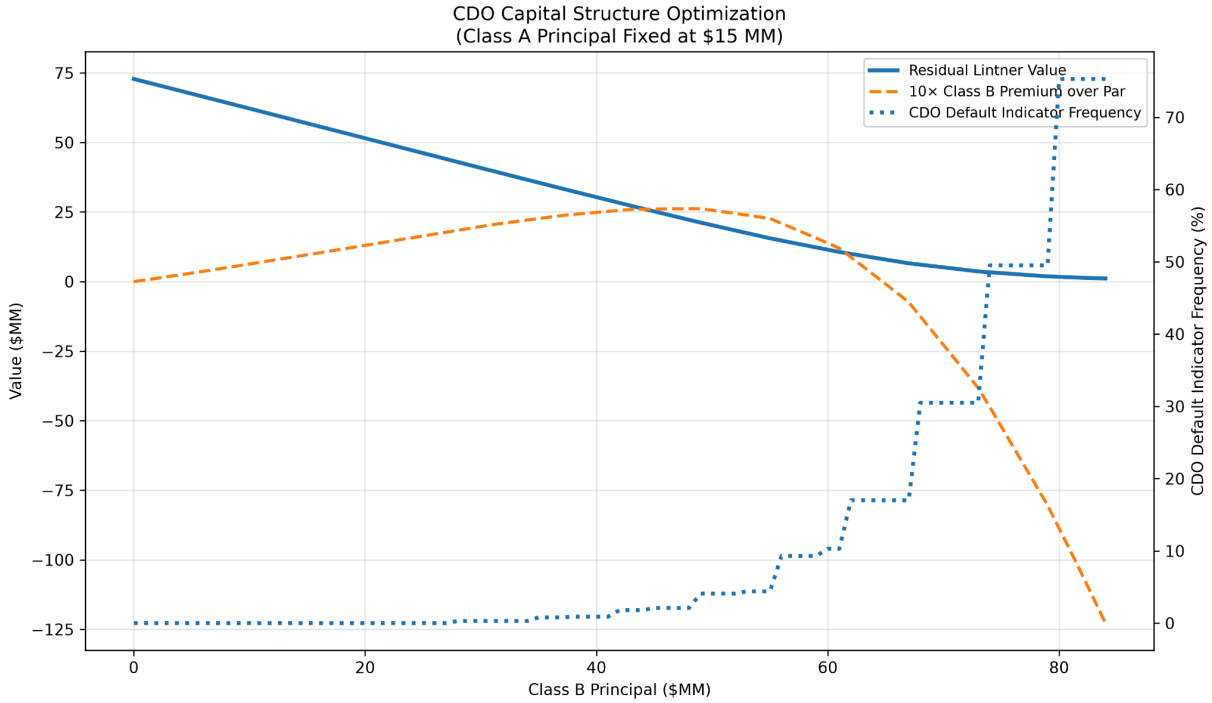


Figure 1. CDO capital structure optimization with Class A principal fixed at $15 MM. The chart shows residual Lintner value, 10× Class B premium over par, and the CDO default-indicator frequency across Class B principal sizes.

Conclusion

The Part II analysis produces a coherent Lintner valuation of the CDO using the original 9% YTM assumption for collateral market value. Moment matching improves the stability of the 1,000-scenario simulation, while the covariance decomposition makes clear that the residual tranche carries almost all of the pool's systematic cash-flow risk. The debt tranches therefore receive only small Lintner risk deductions, whereas the residual absorbs nearly the full aggregate risk charge. The bonus analysis further illustrates the leverage-risk trade-off: increasing Class B principal reduces the residual claim and eventually raises default frequency sharply. Under the fixed 1,000-scenario sample, the maximum Class B principal with zero observed payment defaults is $27.90 MM.